

EEG- and Micro-Expression-Based Emotion Recognition and Consciousness Detection

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Abstract—In the field of Disorders of Consciousness (DoC), traditional diagnostic methods often struggle with high misdiagnosis rates and lack the sensitivity needed to discern subtle emotional responses, necessitating more accurate and comprehensive assessment tools. Current emotion recognition techniques, primarily using single-modality data such as electroencephalography (EEG) or facial expressions, fail to capture the complex interplay of emotional states in patients with DoC, leading to incomplete assessments. To address these challenges, this study proposes a novel multimodal fusion framework that integrates heterogeneous EEG and micro-expression data through an innovative spatio-temporal attention mechanism. This approach significantly enhances the detection and classification of emotional states by leveraging differential entropy features from EEG and temporal features from micro-expressions, fused using deep learning techniques. The proposed method not only achieves mainstream models in terms of accuracy but also demonstrates potential for broad application in medical diagnostics, providing a basis for more personalized patient care and opening avenues for future research in clinical practice settings. We found that two patients had significantly better indicators than the others, indicating better brain activity levels and outcomes in these two patients, which is also consistent with the clinical results one month after the experiment. By effectively merging data from multiple sources, our study sets a new benchmark for emotion recognition in DoC, contributing to more accurate clinical assessments and enhancing the quality of life for affected patients.

Index Terms—Electroencephalogram (EEG), micro-expression, emotion recognition, consciousness detection, disorders of consciousness (DoC).

I. INTRODUCTION

EMOTION is a complex psychological state expressed through diverse physiological and behavioural responses [1]. Emotion recognition is a key research area in psychology and medicine, crucial for understanding individual emotional experiences and assessing states of consciousness, especially in patients with disorders of consciousness (DoC). Over the past few decades, research has explored emotion

expression through facial expressions [2], gestures [3], speech [4], and physiological signals [5]. Physiological signals, such as electromyography (EMG), electrocardiography (ECG), and electroencephalography (EEG), are objective manifestations of emotions that cannot be concealed. Normal facial expressions can be controlled or falsified, concealing genuine emotions, while micro-expressions are brief and subtle changes in expression that occur unconsciously and can serve as indicators of true emotions [6]. EEG, representing brain activity and central nervous system regulation, is difficult to disguise and enables the analysis of underlying psychological activity, effectively complementing facial expression-based emotion recognition.

The three types of DoC include coma, unresponsive wakefulness syndrome (UWS) [7], also known as vegetative state (VS) [8], and minimally conscious state (MCS) [9]. Recent studies have demonstrated that patients with DoC can exhibit neural responses to audiovisual stimuli, even when they are unable to respond behaviorally. For instance, EEG responses to familiar and non-familiar emotional videos can be distinguished in patients with DoC, indicating that they can process such stimuli at a neural level [10]. Similarly, naturalistic movie stimuli have been used to assess awareness in patients with DoC, revealing that a significant number of patients showed neural evidence of conscious awareness [11].

These findings suggest that EEG and micro-expressions can be used to diagnose the residual state of consciousness in patients with DoC. This approach provides a simpler and more effective alternative to traditional motor response-based assessment methods and fMRI.

Despite these advancements, existing solutions for emotion recognition in patients with DoC still face several critical challenges. First, patients with DoC often exhibit unreliable motor-based responses, making it difficult to accurately assess their emotional states. Patients in the UWS state are unable to express intelligible language or open their eyes spontaneously to obey verbal commands [12]. In contrast, those with MCS exhibit inconsistent but reproducible signs of awareness through behavioral responses [9]. This unreliability hinders the accurate assessment of consciousness.

Second, current methods suffer from low sensitivity, leading to high misdiagnosis rates. Motor response-based clinical assessment methods, such as the Coma Recovery Scale-Revised (CRS-R) [13], are used to evaluate the level of consciousness in patients with DoC. However, these methods have high misdiagnosis rates of up to 37%-43% [12]. Functional magnetic resonance imaging (fMRI) has been employed to measure brain activities in patients with DoC, but it is expensive,

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cumbersome, and requires professional expertise to operate and analyze results, making it challenging to implement in clinical settings [14].

Third, current emotion recognition methods often suffer from insufficient cross-modal fusion, limiting their effectiveness in capturing the full spectrum of emotional signals. Existing methods typically focus on single-modal approaches, which are insufficient for capturing the subtle and complex emotional responses in patients with DoC. Even when EEG and facial expressions are jointly utilized, most studies still follow two simplified strategies:

- Decision-level fusion, which merely performs weighted averaging or voting on unimodal predictions and overlooks low-level interactions.
- Early feature-level fusion that simply concatenates features or applies a shared-attention layer after premature fusion, as in HALCB [15], MAMN [16] and HSAttentionNet [17].

These "concatenate-then-share" paradigms suffer from sampling rate mismatch and heterogeneous temporal granularity. They smooth EEG dynamics and dilute expression details, failing to exploit the complementary advantages of the two modalities for weak emotional cues in patients with DoC.

To address these challenges, this study proposes a multimodal emotion recognition approach combining EEG and micro-expressions. We design a spatio-temporal attention mechanism that adaptively weights cross-modal features, thereby enhancing the detection of faint emotional signals in DoC patients. By integrating EEG, a direct reflection of brain activity, and micro-expressions, which carry rich yet subtle information, our method provides a more comprehensive and precise description of the emotional state of patients with DoC. This integration aims to capture the patient's emotional experience at a multimodal level, offering a more convenient method for consciousness detection and a powerful aid for healthcare professionals.

Furthermore, we developed a passive emotion-evoking paradigm and gathered EEG and facial image data from 21 DoC patients (8 UWS patients and 13 MCS patients). Our multimodal feature fusion method combined EEG and micro-expression features, which were then used for emotion classification through deep learning. The evaluation metrics used to assess the emotional expression of DoC patients were Euclidean distance and cosine similarity. We found that 2 UWS patients had significantly better indicators than did the other UWS patients, suggesting that these 2 UWS patients had better levels of brain activity and were able to have a better prognosis. Clinical evaluation one month after the experiment confirmed the accuracy of the assessment. This study explored the application of multimodal emotion recognition in assessing states of consciousness, providing new directions for personalized medicine.

The main contributions of this paper are as follows:

- 1) **Spatio-Temporal Attention Module for Heterogeneous Multimodal Fusion:** We propose a domain-coordinated attention module that adaptively aligns and integrates heterogeneous features across both time and space. This

mechanism is modality-independent and can be seamlessly integrated into various architectures, enhancing unimodal streams (EEG or micro-expressions) or fusing multimodal data without requiring architectural redesign. This approach goes beyond simple concatenation or weighted averaging, providing a more sophisticated and effective method for multimodal fusion.

- 2) **Hybrid Feature Extraction:** The low-dimensional features are extracted to more relevant information by the spatio-temporal attention module, and then the high-dimensional features are generated by the neural network. This hybrid approach enhances the accuracy of emotion recognition, demonstrating the efficacy of integrating both specific low-dimensional and complex high-dimensional features.
- 3) **Clinically Grounded Evaluation Paradigm:** We develop a novel evaluation paradigm that uses emotion-response similarity (via Euclidean distance and cosine similarity between DoC patients and healthy controls) as a proxy for residual consciousness. This paradigm enhances emotion recognition accuracy and informs personalized medical and nursing decisions. Our multimodal emotion recognition framework offers new methods and concepts for clinical practice in consciousness detection.

The remainder of this paper is structured as follows. Section II describes related works. Section III presents the proposed methodology. Section IV describes the experiments, including datasets, data preprocessing, performance evaluation metrics, and experimental settings. Section V presents the experimental results. Finally, Sections VI and VII present the discussion and conclusion, respectively.

II. RELATED WORKS

A. Unimodal Emotion Recognition

In unimodal emotion recognition, researchers have made progress using biometric features like EEG signals and micro-expressions. EEG provides direct brain activity data, while micro-expressions reveal brief emotional changes. These methods have advantages over traditional ones like speech or text analysis, which can be limited by noise or emotion masking.

In EEG-based emotion recognition, deep learning models have become dominant. Song et al. [18] introduced a dynamic adjacency matrix to capture functional connections between EEG channels. Zhang et al. [19] used graph convolution to enhance feature learning, and Chen et al. [20] applied broad learning to integrate multi-level features. These models focus on single tasks and can overfit. To improve generalization, Li et al. [21] added self-supervised learning tasks like spatial puzzles and contrastive learning, reducing the impact of label noise. However, these methods still lack consideration for individual differences, affecting accuracy.

Micro-expressions are extremely short in duration, with low motion amplitude and limited texture changes, making them hard to detect with traditional macro-expression methods. Li et al. [22] used temporal interpolation and motion magnification to enhance detection, but found that longer sequences diluted movement. Liu et al. [23] designed a robust optical

flow method to extract main directional mean optical flow (MDMO), showing that magnitude is more discriminative. Our proposed spatio-temporal attention module focuses on changing regions in video sequences, assigning different weights to capture small facial movements more efficiently.

B. Multimodal Feature Fusion

Multimodal emotion recognition leverages the strengths of multiple modalities (e.g., EEG, EMG, eye movement, and facial expressions) to improve accuracy and robustness. This approach combines information from two or more modalities to make predictions, categorized into decision-level (late) fusion and feature-level (early) fusion.

Decision-level (late) fusion involves making independent predictions for each modality and then integrating these decisions using mechanisms such as weighted sum [24], averaging [25], majority voting [26], or a learnable model [27]. This method improves flexibility but often overlooks low-level interactions among modalities, potentially harming performance.

Feature-level (early) fusion extracts separate features from each modality and then combines them to construct a joint representation. The most common early fusion technique is to concatenate the features from each modality to construct a multimodal joint representation. Recent advances in deep learning have enabled more sophisticated fusion techniques that exploit the complementarity of multimodal data [28]. Among them, attention-based fusion has become the dominant paradigm because it can dynamically re-weight modality contributions.

Representative methods include HALCB [15] and MAMN [16], which use single-shared attention layers to refine the joint representation. However, these methods apply a shared attention module after simple concatenation, ignoring modality-specific characteristics. Liu et al. [29] further introduce polarized self-attention (PSA) to suppress redundant cross-modal information, yet PSA still operates on a homogeneous feature space after early concatenation. HSAttentionNet [17] employs parallel attention heads for each modality, but the features are immediately fused via element-wise addition, which may blur the temporal granularity of low-sampling-rate signals such as EEG.

In summary, existing attention-based fusion strategies either apply a shared attention module after simple concatenation, ignoring modality-specific characteristics, or perform attention after premature fusion, risking temporal misalignment and neurophysiological detail loss. In contrast, our approach applies modality-specific spatio-temporal attention before lightweight cross-modal alignment. This design preserves the distinct space-time structure of EEG and micro-expressions, ensuring that the unique characteristics of each modality are fully leveraged. By doing so, our method avoids the pitfalls of both sampling rate mismatch and loss of neurophysiological details, thereby providing a more accurate and robust fusion framework.

C. Consciousness Detection in Patients with DoC

The clinical diagnosis of patients with DoC is typically based on behavioral scales like the Coma Recovery Scale-

Revised (CRS-R), which rely on observable motor responses [30]. However, these responses can be difficult to detect or inconsistent, leading to potential misdiagnosis [31]. To address this, researchers have turned to EEG and fMRI to identify residual brain functions and detect consciousness independent of motor abilities [14, 32, 33].

BCIs translate brain activity into control signals, enabling communication for those with motor disabilities [34]. BCIs can also explore residual cognitive functions, such as emotion recognition, in DoC patients. Recent studies using BCI paradigms have shown promise [33, 35, 36]. For example, Lulé et al. [37] used an auditory P300-based BCI to detect consciousness in patients with DoC, while Pan et al. [38] developed a visual hybrid BCI combining P300 and SSVEPs. However, due to the severe brain injuries suffered by patients with DoC, there are large differences in EEG signals between these patients and healthy individuals. Thus, researchers have aimed to develop new BCIs to enhance the accuracy of consciousness detection.

Recent studies show that emotion elicitation techniques can enhance BCI accuracy. Mirror Neuron System (MNS)-based techniques, which evoke brain activity through observing emotional facial expressions, are widely used [39]. Research on EEG-based emotion recognition in healthy individuals has also advanced, with studies achieving high accuracy using various methods [40–42]. However, emotion recognition in DoC patients remains underexplored. Emotion recognition is a critical cognitive function linked to attention, memory, language, and decision-making [43, 44]. It is used to evaluate cognitive impairments in conditions like schizophrenia, ADHD, and Parkinson's disease [45–47]. While not included in clinical scales like the CRS-R, emotion recognition tasks could help assess residual cognitive functions in DoC patients. A passive emotion-evoking paradigm can determine awareness levels based on emotional changes, supplementing CRS-R assessments and improving the evaluation of consciousness in DoC patients.

III. METHODOLOGY

A. Overview of the Proposed Framework

Figure 1 illustrates our foundational fusion framework for emotion recognition, which addresses the challenge of heterogeneous multimodal integration through a spatio-temporal attention mechanism. The process begins with feature extraction: MTCNN [48] selects facial images, and raw EEG data is preprocessed. These features are fed into the emotion recognition model, which includes a spatiotemporal attention encoder (STAE) to fuse multimodal data. We use Swin-T [49] as the backbone network. Emotions are classified by an MLP layer, which outputs probabilities for positive, neutral, or negative emotions. For consciousness detection, we compute distance and similarity metrics from the classified emotions of DoC patients and healthy subjects. This framework integrates image processing and neural networks to accurately detect emotions and consciousness by analyzing facial expressions and brainwave patterns. Details are provided in the following sections.

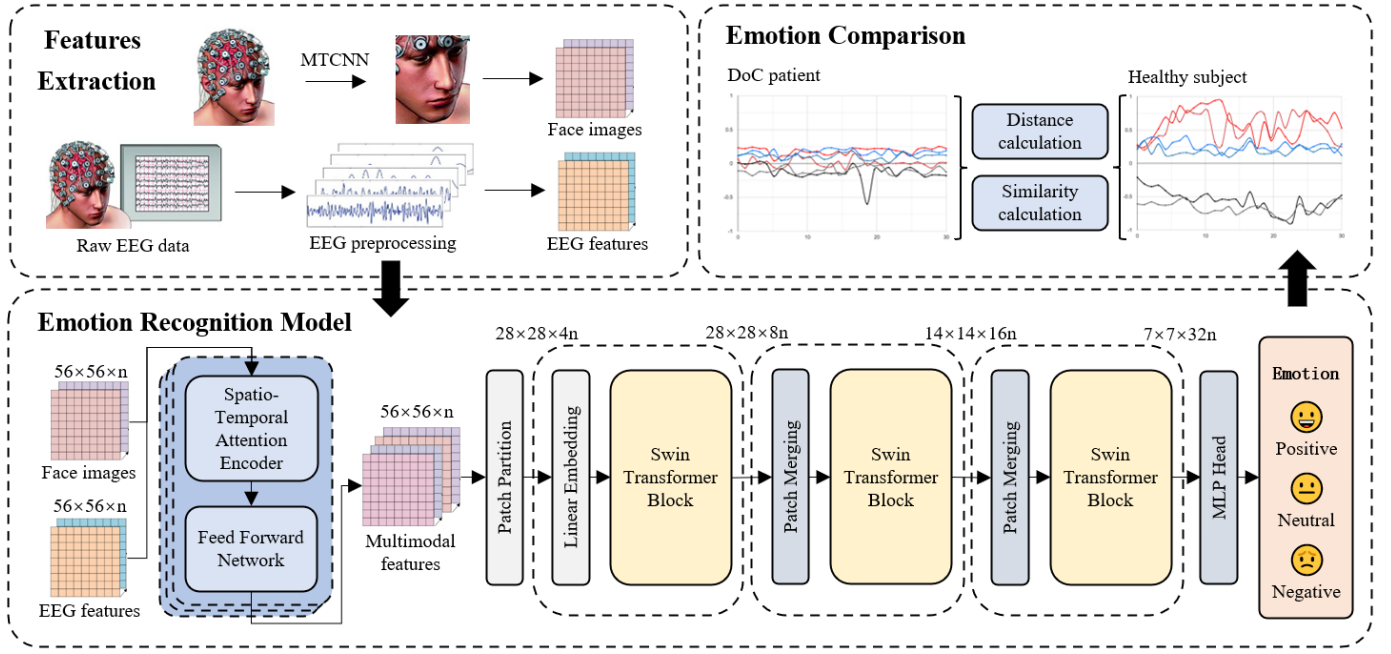


Fig. 1. Proposed STST framework for simultaneous emotion recognition and consciousness detection in patients with Disorders of Consciousness (DoC).

- ① **Input:** 32-ch EEG (0.1–50 Hz) and 5-frame micro-expression sequences (≈ 0.2 s) are acquired during an audiovisual emotion-elicitation paradigm.
- ② **Feature extraction:** MTCNN performs face detection & alignment; DE features from 5 EEG bands (δ – γ) and $56 \times 56 \times 5$ face tensors are simultaneously fed into the spatio-temporal attention encoder (STAE). Low-dimensional handcrafted DE features and high-dimensional tensors are jointly extracted, providing richer emotion-related cues than either stream alone.
- ③ **Multimodal fusion & classification:** Attention-weighted multimodal tokens are concatenated and processed by a Swin-Transformer backbone; an MLP head outputs discrete emotion probabilities (positive, neutral, negative). Modality-independent attention module requires no architectural redesign when either modality is missing, outperforming simple concatenation or weighted averaging.
- ④ **Consciousness paradigm:** Emotion vectors of DoC patients are compared with those of healthy controls via Euclidean distance and cosine similarity; smaller distance & larger similarity indicate higher residual consciousness. This paradigm maps the metrics directly to CRS-R-guided consciousness levels for personalized clinical decisions.

B. Feature Extraction

The attention mechanism is commonly used in visual and natural language processing tasks to locate the correct regions for image captioning or to concentrate on the key part of a sentence given the aspect. In this study, we propose a spatio-temporal attention module to focus on the associations of multi-channel EEG signals and extract micro-expression features. Finally, we perform feature fusion of the EEGs and micro-expressions at the feature level to form multimodal features.

EEG Feature The raw EEG signal is first preprocessed, which includes artifact rejection, filtering, and component removal. After preprocessing, the EEG data from each channel are segmented into non-overlapping 1-second intervals based on the output size to extract EEG features. Previous studies have confirmed that in emotion recognition tasks, DE outperforms ES, DASM, RASM, and ASM [50]. The EEG signal after the bandpass filter obeys a Gaussian distribution $N(\mu, \sigma^2)$, and the DE feature of the EEG is calculated as follows:

$$\begin{aligned} DE &= - \int f(x) \log[f(x)] dx \\ &= - \int \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \log \left[\frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \right] dx \quad (1) \\ &= \frac{1}{2} \log(2\pi e \sigma^2) \end{aligned}$$

where $f(x)$ is the probability density function of x and σ^2 is the variance of this segment of the EEG.

The DE feature belongs to the category of frequency domain attributes. Typically, EEG signals for emotion recognition are divided into five different frequency bands: δ (0.1–4 Hz), θ (4–8 Hz), α (8–12 Hz), β (12–30 Hz), and γ (30–45 Hz). In our study, we computed the DE feature for each frequency band separately. The final feature vector for each sample contains 56 channels (or 28 channels $\times$ 2) $\times$ 56 DE features $\times$ 5 frequency bands to ensure that the sizes of the micro-expression features match.

Micro-expression Feature Micro-expressions are brief, involuntary facial expressions that reveal concealed emotions. In recent years, there have been significant advancements in feature extraction techniques for micro-expressions. Liu et al. [51] introduced a hybrid approach that combines man-

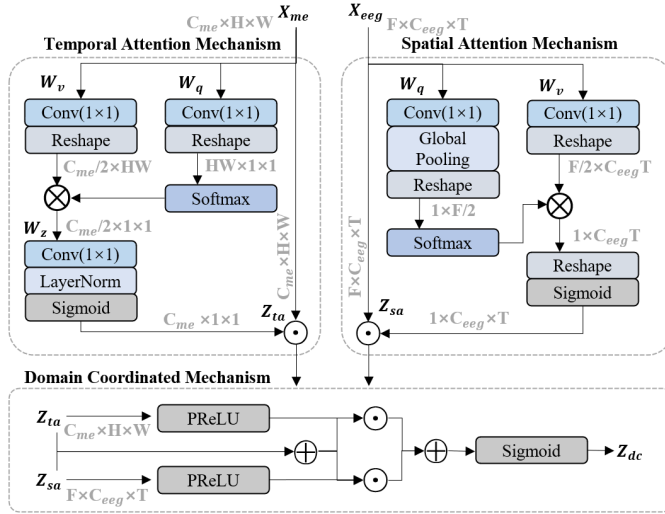


Fig. 2. Feature fusion architecture using spatiotemporal attention mechanisms for EEG and micro-expression data.

ual AU coding with automated 3D facial feature analysis, providing a richer emotional context. Similarly, Zhi et al. [52] developed an improved Facial Action Coding System (FACS) that utilizes machine learning algorithms to identify facial action units (AUs) more efficiently. In our study, we aimed to combine micro-expression and EEG modal features. To achieve this goal, we used a novel method for extracting and annotating micro-expression features. The MTCNN is used for face detection and alignment during the raw image processing of the exported video frames. To ensure efficient processing, we selected sequences of five facial images that were evenly distributed over the entire duration of the micro-expression (typically lasting from 1/25 to 1/5 of a second). The purpose of this selection process is used to capture the fundamental dynamics of micro-expressions within the limited timeframe of a micro-expression. This process was guided by the definition of micro-expressions as brief, involuntary facial expressions that reveal concealed emotions. We resized each image to a consistent 56×56 pixel format and stacked them to form a $5 \times 56 \times 56$ feature tensor. We extracted micro-expression features from the identified variations using the spatio-temporal attention mechanism, which were then utilized for fusion with EEG data. This tensor captures the fundamental dynamics of facial expressions within the limited timeframe of a micro-expression, ensuring that crucial information regarding micro-expressions is retained for feature fusion and subsequent analysis.

Fusion of EEG and Micro-expression Features

We present a modality-respecting spatio-temporal attention mechanism specifically engineered for the heterogeneous EEG–micro-expression pair. Unlike generic spatio-temporal attention, our architecture first preserves the intrinsic space-time structure of each modality. Spatial attention only re-weights EEG channels/frequency bands, while temporal attention only highlights micro-expression frames before any cross-modal fusion. This alignment-first design explicitly exploits the neural versus facial complementary characteristics and

constitutes the core technical novelty of our approach, as illustrated in Figure 2.

To explicitly address the heterogeneity between EEG and micro-expression modalities, we devise a two-stage fusion strategy. (1) Modality-specific preprocessing: handcrafted features from both modalities are individually z-score normalized so that each channel or image sequence has zero mean and unit variance, guaranteeing that the disparate numerical ranges do not bias subsequent attention weighting. (2) Semantic-level fusion: the normalized tensors are fed into the proposed modality-respecting spatio-temporal attention encoder, which dynamically generates modality-aware importance maps and outputs a unified representation. Formally, let the EEG tensor be $X_{\text{eeeg}} \in \mathbb{R}^{F \times C_{\text{eeeg}} \times T}$ and the micro-expression tensor be $X_{\text{me}} \in \mathbb{R}^{C_{\text{me}} \times H \times W}$. The spatial attention module produces the EEG spatial map Z_{sa} , while the temporal attention module yields the micro-expression temporal map Z_{ta} . These two heterogeneous but complementary maps are then fused through the domain-coordinated mechanism.

In the temporal attention module, the input micro-expression features X_{me} are initially processed through a 1×1 convolutional layer denoted as W_v to generate a transformed feature:

$$V_{\text{me}} = W_v(X_{\text{me}}) \in \mathbb{R}^{C_{\text{me}}/2 \times H \times W} \quad (2)$$

Subsequently, these features undergo reshaping to a dimensionality of $C_{\text{me}}/2 \times HW$, with H and W symbolizing the height and width of the feature maps, respectively:

$$V_{\text{me}}' = \sigma_1(V_{\text{me}}) \in \mathbb{R}^{C_{\text{me}}/2 \times HW} \quad (3)$$

This 1×1 convolution keeps the 56×56 spatial size intact while halving the channel dimension, preparing micro-expression features for temporal-only importance scoring and preserving modality-specific semantics. Another 1×1 convolutional layer, W_q , is employed to produce query features:

$$Q_{\text{me}} = W_q(X_{\text{me}}) \in \mathbb{R}^{1 \times H \times W} \quad (4)$$

These query features are then passed through a softmax layer to compute an attention distribution:

$$Q_{\text{me}}' = F_{\text{sm}}(\sigma_2(Q_{\text{me}})) \in \mathbb{R}^{HW \times 1 \times 1} \quad (5)$$

This attention map is element-wise multiplied with the transposed features to yield the temporal-only attention feature map Z_{ta} :

$$Z_{\text{ta}} = F_{W_z}(V_{\text{me}}' \otimes Q_{\text{me}}') \odot X_{\text{me}} \in \mathbb{R}^{C_{\text{me}} \times H \times W} \quad (6)$$

where W_v , W_q and W_z are 1×1 convolution layers, σ_1 and σ_2 are two tensor reshape operators, $\otimes$ is the matrix dot product operation, and $\odot$ is the channel-wise multiplication operator. F_{sm} is the Softmax operator. F_{W_z} consists of W_z convolutional operations, followed by the LayerNorm and Sigmoid operators, so that only frame-wise importance is modulated while the spatial structure remains intact.

In the spatial attention mechanism, the input EEG features X_{eeeg} are initially processed through a 1×1 convolutional layer denoted as W_v to generate a transformed feature:

$$V_{\text{eeeg}} = W_v(X_{\text{eeeg}}) \in \mathbb{R}^{F/2 \times C_{\text{eeeg}} \times T} \quad (7)$$

Subsequently, these features undergo reshaping to a dimensionality of $F/2 \times C_{\text{eeg}}T$, with F and T symbolizing the frequency bands and time dimension, respectively:

$$\mathbf{V}_{\text{eeg}}' = \sigma_1(\mathbf{V}_{\text{eeg}}) \in \mathbb{R}^{F/2 \times C_{\text{eeg}}T} \quad (8)$$

Another 1×1 convolutional layer, W_q , is employed to produce query features. These query features are then passed through a global pooling layer to compute a pooled feature:

$$\mathbf{Q}_{\text{eeg}} = G(W_q(X_{\text{eeg}})) \in \mathbb{R}^{F/2 \times 1 \times 1} \quad (9)$$

These features are then passed through a softmax layer to compute an attention distribution:

$$\mathbf{Q}_{\text{eeg}}' = F_{\text{sm}}(\sigma_2(\mathbf{Q}_{\text{eeg}})) \in \mathbb{R}^{1 \times F/2} \quad (10)$$

This attention map is element-wise multiplied with the transposed features to yield the spatial-only attention feature map Z_{sa} :

$$Z_{\text{sa}} = \sigma_s(\mathbf{Q}_{\text{eeg}}' \otimes \mathbf{V}_{\text{eeg}}') \odot X_{\text{eeg}} \in \mathbb{R}^{F \times C_{\text{eeg}} \times T} \quad (11)$$

where W_q and W_v are 1×1 convolution layers, σ_1 and σ_2 are two tensor reshape operators, $\otimes$ is the matrix dot product operation, and $\odot$ is the channel-wise multiplication operator. G is the Global Pooling operator. F_{sm} is the Softmax operator. σ_s consists of a tensor reshape operator, followed by a Sigmoid operator, so that only channel-wise importance is modulated while temporal dynamics remain intact.

In the domain coordination mechanism, the temporal attention feature Z_{ta} and the spatial attention feature Z_{sa} are nonlinearly transformed by PReLU and combined through a gated-residual fusion. This step performs cross-modal alignment after each modality has been individually optimized for its own space-time structure, preventing the heterogeneity from being diluted by premature mixing. The calculation formula for the domain coordinated mechanism is as follows:

$$Z_{\text{dc}} = F_{\text{sg}}((F_{\text{PR}}(Z_{\text{ta}}) + F_{\text{PR}}(Z_{\text{sa}})) \odot (Z_{\text{ta}} + Z_{\text{sa}})) \quad (12)$$

where Z_{dc} represents the output obtained when the inputs are the two dimensions of EEG and micro-expression. $\odot$ is the channel-wise multiplication operator. F_{sg} represents the sigmoid operator, and F_{PR} represents the PReLU operator.

In the fusion model for EEG and micro-expression features, as depicted in Appendix A, the attention mechanism is consolidated to accommodate a single modality input X . The input X can be a single modality, such as an EEG feature, a micro-expression feature, or a feature that has been fused. In our model, we first combined the EEG and micro-expression features using the structure shown in Figure 2. To enhance the feature characterization, we then overlay multiple layers of unimodal inputs onto the spatio-temporal attention mechanism.

In summary, the proposed approach is not a re-hash of generic spatio-temporal attention; it is a modality-respecting, alignment-first framework that isolates the distinct space-time structure of neural and facial signals, generates complementary attention maps with minimal parameters, and fuses them through a gated mechanism that preserves modality-specific emotional cues.

C. Emotion Recognition Model

In the field of multimodal emotion recognition, various backbone networks have been explored, including CNNs for spatial features and RNNs (e.g., LSTMs) for temporal features. Furthermore, recent studies have shown that hybrid models that combine CNNs and RNNs can effectively capture both spatial and temporal features [53].

We developed the Spatiotemporal Swin Transformer (STST) model, leveraging the Swin Transformer architecture for its superior performance in vision tasks. "Spatiotemporal" indicates our focus on both spatial and temporal aspects of the multimodal data, which are essential for capturing the dynamics of emotional states in EEG and micro-expression analysis. The core innovation lies in the STAE layers that harmonizes the heterogeneous EEG and micro-expression tensors into a unified representation before any Swin Transformer layer is applied. Consequently, the subsequent self-attention modules operate on a already-aligned multimodal manifold, rather than on raw concatenated features, which significantly reduces the risk of modality dominance and over-fitting on small clinical datasets.

In our study, we developed the Spatiotemporal Swin Transformer (STST) model. The STST name aptly reflects our model's core strengths and design ethos. The term "spatiotemporal" signifies our comprehensive consideration of both spatial and temporal dimensions within multimodal data, which is crucial for accurately capturing the dynamic characteristics of emotional states in EEG and micro-expression analysis. The proposed spatiotemporal attention mechanism is designed to effectively fuse heterogeneous bimodal features from EEG and micro-expressions. This mechanism dynamically adjusts the weights of different features based on their importance, thereby enhancing the representation of significant features. Specifically, it captures both temporal and spatial information within the features, identifies correlations in EEG signals across different time points and frequency bands, and detects subtle changes in micro-expressions over short durations. The "Swin Transformer" underscores the model's foundation for cutting-edge Swin Transformer architecture, which is renowned for its exceptional performance in various vision tasks. The Swin Transformer was selected as the backbone network for our model due to its superior performance in various vision-related tasks, adaptability, and scalability in processing multimodal data. When customizing the base variant of the Swin Transformer, we made minor yet significant modifications to better accommodate our multimodal data. Specifically, we optimized the patch size and network depth to align with the dimensions of our feature tensor. This ensures that our STST model is tailored to the requirements of emotion recognition tasks and can leverage the full potential of the Swin Transformer architecture.

The architecture of the backbone network is based on Swin-T. The multimodal feature tensor, with dimensions of $56 \times 56 \times n$, enters a patch partitioning stage where it is divided into non-overlapping patches. These patches are then linearly embedded to transform the patch-separated features into tokens suitable for the Transformer's input. Subsequently, these

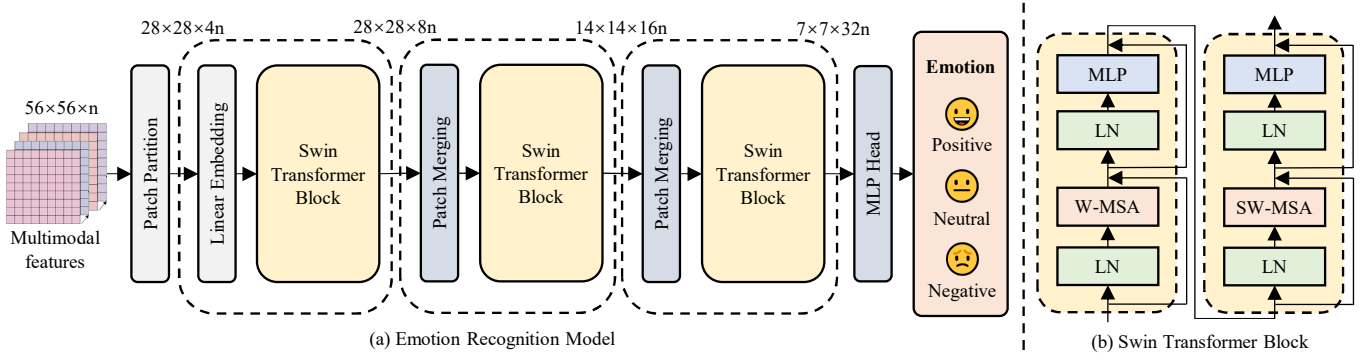


Fig. 3. Overview of the Emotion Recognition Model (a) and Swin Transformer Block (b).

tokens pass through a series of Swin Transformer blocks that progressively merge and refine the feature maps. Specifically, it captures both temporal and spatial information within the features, identifies correlations in EEG signals across different time points and frequency bands, and detects subtle changes in micro-expressions over short durations. To provide a clearer understanding of the Swin Transformer block's architecture, we include Figure 3 below. This diagram illustrates the detailed structure of the Swin Transformer block, highlighting the key components. W-MSA and SW-MSA are multi-head self attention modules with regular and shifted windowing configurations, respectively. Each Swin block consists of multiple layers of self-attention mechanisms and multilayer perceptrons (MLPs), with layer normalization (LN) applied before each block and residual connections after each. They enable the model to capture both local and global contextual information, which is essential for accurately recognizing emotions from multimodal data. By leveraging the hierarchical processing of the Swin Transformer architecture, our model can effectively integrate spatiotemporal information from facial images and EEG data, leading to a semantically rich feature set that is highly discriminative for emotion recognition. The MLP consolidates the features into probability scores corresponding to the emotional categories of positive, neutral, and negative. This end-to-end trainable system ensures that the classification process leverages both the spatiotemporal patterns and the refined features obtained from the Swin Transformer blocks, culminating in a robust emotion recognition model capable of high accuracy.

D. Emotion Comparison

We validate our method on public datasets using accuracy as a benchmark. For our self-recorded dataset, we first test the model on 10 healthy subjects to establish a baseline. We then apply it to 21 patients, mapping predicted probabilities to continuous emotion scores (-1 to 1) using a weighted approach, where -1 is negative, 0 is neutral, and 1 is positive. The formula representing this process is as follows:

$$\text{Score} = C_{\text{neg}} \times P_{\text{neg}} + C_{\text{neu}} \times P_{\text{neu}} + C_{\text{pos}} \times P_{\text{pos}} \quad (13)$$

where C_{neg} , C_{neu} , and C_{pos} represent the contribution values associated with negative, neutral, and positive emotional states,

respectively, assigned as -1, 0, and 1. P_{neg} , P_{neu} , and P_{pos} represent the probabilities predicted by the deep learning model for negative, neutral, and positive emotions, respectively. This formula aims to transform the probabilistic outputs of emotion classification into a continuous score ranging from -1 to 1, thereby providing a more accurate composite assessment of emotional tendencies. To evaluate the model's output, we use the Euclidean distance and cosine similarity as metrics for the distance calculation and similarity assessment, respectively. These metrics measure the degree of similarity between the emotional patterns of patients with DoC and healthy subjects.

Emotions are closely linked to consciousness, making emotional state analysis useful for evaluating consciousness levels in DoC patients. We correlate changes in CRS-R scores with emotion recognition metrics to explore associations between emotional variability and consciousness levels. It can alert patients whose emotion recognition metrics show significant variances, which can facilitate better prognoses and potentially expedite rehabilitation processes.

IV. EXPERIMENTS

A. Datasets

1) *Public Datasets*: Our study included various public datasets that cover multiple modalities, including the EEG datasets SEED [41] and SEED-IV [54], the micro-expression datasets CASME II [55], SAMM [56] and SMIC [57], and the multimodal dataset MAHNOB-HCI [58].

- **EEG**: SEED (15 subjects, 45 trials, 62-ch, 1000 Hz $\rightarrow$ 200 Hz, 3-class) and SEED-IV (15 subjects, 45 trials, 62-ch, 1000 Hz $\rightarrow$ 200 Hz, 4-class).
- **Micro-expression**: CASME II (26 subjects, 247 clips, 200 fps, 5-class), SAMM (32 subjects, 159 clips, 200 fps, 8-class) and SMIC (16 subjects, 164 clips, 100 fps, 3-class).
- **Multimodal**: MAHNOB-HCI (27 subjects, 32-ch 256Hz EEG + 60fps video).

2) *Self-recorded Dataset*: Our self-recorded dataset included EEG signals and facial expression images. To collect high-quality scalp EEG signals, 32-electrode scalp EEG signals were recorded using a NuAmps amplifier (Neuroscan Compumedics, USA) and an EEG cap. The impedance of all 32 electrodes was kept below 30 k Ω . The EEG signals were

TABLE I
OVERVIEW OF CLINICAL DIAGNOSIS PATIENTS

Patient	Clinical diagnosis	Etiology	Time since injury (months)	CRS-R Score		
				One week before experiment	One month after experiment	Three months after experiment
P1	MCS	NTBI	7	(16) 3-4-5-2-0-2 (MCS)	(17) 3-4-6-2-0-2 (eMCS)*	Rehabilitation*
P2	UWS	HBI	8	(9) 2-1-2-2-0-2 (UWS)	(9) 2-1-2-2-0-2 (UWS)	(9) 2-1-2-2-0-2 (UWS)
P3	MCS	HBI	2	(13) 3-1-5-2-0-2 (MCS)	Rehabilitation*	Rehabilitation*
P4	UWS	HBI	17	(6) 1-0-2-1-0-2 (UWS)	(6) 1-0-2-1-0-2 (UWS)	(6) 1-0-2-1-0-2 (UWS)
P5	UWS	HBI	8	(8) 2-1-2-1-0-2 (UWS)	(8) 2-1-2-1-0-2 (UWS)	(8) 2-1-2-1-0-2 (UWS)
P6	MCS	NTBI	3	(12) 1-3-5-1-0-2 (MCS)	(12) 1-3-5-1-0-2 (MCS)	(12) 1-3-5-1-0-2 (MCS)
P7	MCS	NTBI	2	(11) 0-3-5-1-0-2 (MCS)	(12) 1-3-5-1-0-2 (MCS)†	(19) 3-4-6-3-1-2 (eMCS)*
P8	MCS	TBI	7	(9) 1-3-2-1-0-2 (MCS)	(9) 1-3-2-1-0-2 (MCS)	(9) 1-3-2-1-0-2 (MCS)
P9	UWS	HBI	6	(7) 1-1-2-1-0-2 (UWS)	(9) 1-3-2-1-0-2 (MCS)*	Not Available
P10	MCS	NTBI	2	(14) 2-4-5-1-0-2 (MCS)	(17) 2-5-6-2-0-2 (eMCS)*	Rehabilitation*
P11	MCS	NTBI	5	(14) 4-5-2-1-0-2 (MCS)	Not Available	Not Available
P12	UWS	HBI	9	(7) 1-1-2-1-0-2 (UWS)	(8) 2-1-2-1-0-2 (UWS)†	(8) 2-1-2-1-0-2 (UWS)†
P13	MCS	TBI	3	(9) 1-3-2-1-0-2 (MCS)	(13) 1-3-5-2-0-2 (MCS)†	(23) 4-5-6-3-2-3 (eMCS)*
P14	UWS	NTBI	5	(5) 0-1-2-1-0-1 (UWS)	Not Available	Not Available
P15	MCS	NTBI	4	(9) 1-3-2-1-0-2 (MCS)	(9) 1-3-2-1-0-2 (MCS)	(9) 1-3-2-1-0-2 (MCS)
P16	MCS	TBI	3	(11) 1-2-5-1-0-2 (MCS)	(16) 2-5-6-1-0-2 (eMCS)*	Rehabilitation*
P17	UWS	NTBI	24	(5) 0-1-1-1-0-2 (UWS)	(5) 0-1-1-1-0-2 (UWS)	Not Available
P18	MCS	NTBI	4	(11) 2-3-3-1-0-2 (MCS)	(11) 2-3-3-1-0-2 (MCS)	(11) 2-3-3-1-0-2 (MCS)
P19	UWS	HBI	3	(6) 1-0-2-1-0-2 (UWS)	(8) 1-1-2-2-0-2 (UWS)†	(8) 1-1-2-2-0-2 (UWS)†
P20	MCS	NTBI	2	(6) 0-1-3-1-0-1 (MCS)	(12) 3-1-5-1-0-2 (MCS)†	(12) 3-1-5-1-0-2 (MCS)†
P21	MCS	TBI	3	(8) 0-3-2-1-0-2 (MCS)	(8) 0-3-2-1-0-2 (MCS)	(8) 0-3-2-1-0-2 (MCS)

MCS, Minimally Conscious State; eMCS, emergence from Minimally Conscious State; UWS, Unresponsive Wakefulness Syndrome; TBI, Traumatic Brain Injury; NTBI, Nontraumatic Brain Injury; HBI, Hypoxia-Ischemia Brain Injury; CRS-R: Auditory-Visual-Motor-Oromotor/Verbal-Communication-Arousal;

*: An upgrade in their level of consciousness in diagnosis after experimentation;

†: Improvement in CRS-R score without an upgrade in their level of consciousness in diagnosis after experimentation; Some information, such as age and gender, was hidden to protect the privacy of each patient.

amplified at a sampling rate of 1000 Hz and then bandpass filtered (0.1-50 Hz). The facial expression images were exported as images from a video recorded by the camera at 1080p and 30 fps. The subjects were in a controlled environment with minimal external stimuli. The experimental design included exposure to emotionally arousing stimuli, including images and audio clips.

There are 31 subjects in this dataset. The study included 10 healthy subjects from South China Normal University and 21 patients with DoC in cooperation with Zhujiang Hospital of Southern Medical University. Among the patients with DoC, 13 patients with MCS and 8 patients with UWS were included, as shown in Table I. The experiment was approved by the Ethics Committee of Zhujiang Hospital (approval number: 2023-KY-174-01) and complied with the Code of Ethics of the World Medical Association (Declaration of Helsinki). Prior to the start of the experiment, the patient's family members were informed of the experimental tasks, and written informed consent was obtained from them.

In our experiment, we used a series of emotion-evoking video clips to stimulate the participants. These video clips were carefully selected to include both visual and auditory elements to provide a multimodal sensory experience. Research has shown that combining visual and auditory elements can enhance neural responses in patients with DoC, providing a more comprehensive assessment of their awareness [59]. Our choice of video stimuli aligns with recent studies emphasizing the importance of multimodal sensory experiences in assessing consciousness. The video clips were categorized

into three types of emotions: positive, neutral, and negative. To ensure the effectiveness of emotional arousal in our study, we carefully selected and validated our video stimuli based on established principles and empirical evidence. First, we adopted the selection criteria from the SEED dataset, which has been widely used and validated for eliciting emotional responses. Additionally, we conducted a pre-study validation process with a group of healthy participants. They were shown the same video stimuli, and their emotional responses were assessed through self-reports and physiological measures. This step ensured that our materials effectively elicited the intended emotional responses. The specific videos used in the experiment are listed in Table II, and the experimental paradigm is shown in Figure 4. This multimodal approach was chosen to enhance the overall sensory experience for the participants and to encourage a more robust response to the stimuli.

B. Data Preprocessing

In emotion recognition, data preprocessing is a crucial step. It is directly related to the effect of feature extraction and emotion recognition. In our study, we used both public and self-recorded datasets. We maintained a consistent data preprocessing process for both to ensure comparable and robust results.

For EEG signals, we applied a 0.1–60 Hz bandpass filter and a 50 Hz notch filter to remove noise. We checked for bad leads and interpolated their values using surrounding leads. Independent component analysis was used to remove EOG and

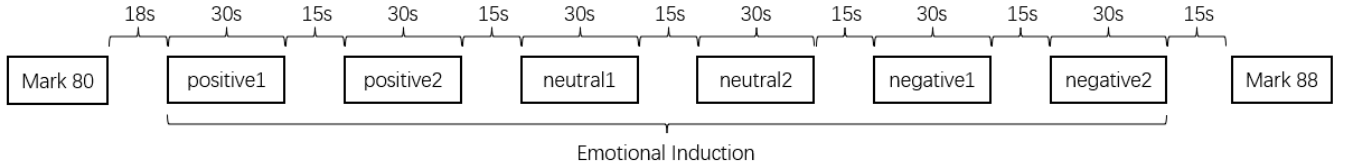


Fig. 4. Flowchart of the emotion elicitation video paradigm.

TABLE II
DETAILS OF FILM CLIPS USED IN OUR EMOTION EXPERIMENT

No.	Emotion label	Film clips sources
1	Positive	"Flirting Scholar"
2	Positive	"Just Another Pandora's Box"
3	Neutral	"Aerial China" Episode 1
4	Neutral	"Aerial China" Episode 2
5	Negative	"Back to 1942"
6	Negative	"Tangshan Earthquake"

Video Format: H.264 codec at 5 Mbps, 1080p resolution, 24 frames per second;
Audio Format: The AAC codec at 192 kbps stereo audio, 48 kHz sampling rate, with a uniform audio gain of -6 dB.

EMG artifacts. The signals were then standardized to ensure consistent amplitude across channels. These preprocessing steps provide a solid foundation for artificial feature extraction and emotion recognition.

For facial images, we loaded the images, converted them to grayscale, and applied histogram equalization to enhance contrast. Faces were detected using MTCNN, and the detected faces were aligned to ensure consistent positions and sizes. We standardized the images so that pixel values have a mean of 0 and a variance of 1.

C. Performance Evaluation Metrics

To comprehensively assess the classification performance of our emotion recognition model, we employed three key metrics: accuracy, Unweighted F1-score (UF1), and Unweighted Average Recall (UAR). These metrics were chosen to provide a balanced evaluation, especially considering the potential class imbalance in our datasets. They are defined as follows:

$$\text{For each class } i: \quad P_i = \frac{TP_i}{TP_i + FP_i}, \quad R_i = \frac{TP_i}{TP_i + FN_i}$$

$$UF1 = \frac{1}{C} \sum_{i=1}^C \frac{2P_i R_i}{P_i + R_i} \quad (14)$$

$$UAR = \frac{1}{C} \sum_{i=1}^C R_i \quad (15)$$

where C is the number of classes, and TP_i , FP_i , FN_i are the true-positive, false-positive, and false-negative counts for class i , respectively.

In emotion recognition for patients with DoC, we cannot directly use the emotion labels of patients with DoC. We cannot ensure that the patient has been induced with the corresponding emotion, so traditional accuracy assessment methods

are not applicable. Instead, we used Euclidean distance and cosine similarity as metrics to quantify the similarity of emotional expressions between healthy subjects and patients with DoC. First, we used Euclidean distance to measure the spatial difference between emotional expressions. For each sample, we represented its emotion score as a vector and calculated the Euclidean distance to quantify the differences between samples. The smaller the Euclidean distance, the closer the emotional intensity is. Second, we used cosine similarity to measure the angular relationship between sentiment vectors. Specifically, the closer the cosine similarity value is to 1, the more similar the trend of sentiment change between the two vectors. Conversely, the closer the value is to -1, the more opposite the trend of sentiment change. For the vector $\mathbf{A} = (a_1, a_2, \dots, a_n)$ of emotion scores of healthy subjects at the time of the evoked video playback and the vector $\mathbf{B} = (b_1, b_2, \dots, b_n)$ of emotion scores of patients with DoC, the Euclidean distance d and the cosine similarity s between them were calculated by the following formulas:

$$d(\mathbf{A}, \mathbf{B}) = \sqrt{\sum_{i=1}^n (a_i - b_i)^2} \quad (16)$$

$$s(\mathbf{A}, \mathbf{B}) = \frac{\sum_{i=1}^n a_i \cdot b_i}{\sqrt{\sum_{i=1}^n a_i^2} \cdot \sqrt{\sum_{i=1}^n b_i^2}} \quad (17)$$

The Euclidean distance is the square root of the sum of the squared differences between the corresponding components of two vectors. The Euclidean distance takes values $[0, 2]$, as we set negative sentiment to -1, neutral sentiment to 0, and positive sentiment to 1. The cosine similarity takes values between $[-1, 1]$, where a value closer to 1 means that the two vectors are more similar, and a value closer to -1 means that the two vectors are more opposite.

D. Experimental Settings

To ensure consistency and comparability with previous studies, we adapted our model's multi-classification strategy by customizing the final MLP Head to align with the specific emotion classification tasks for each dataset. This approach allows our model to be flexibly applied to various emotion recognition tasks while maintaining consistency with existing benchmarks.

- **SEED Dataset:** We configured the model for 3-class emotion classification (positive, neutral, and negative).
- **SEED-IV Dataset:** We set up the model for 4-class emotion classification (happy, sad, afraid, and neutral).

- **Combined CASME II–SAMM–SMIC Dataset:** Following the guidelines of the Micro-Expression Grand Challenge 2019 (MEGC2019), we configured the model for 3-class micro-expression classification (positive, negative, and surprise).
- **MAHNOB-HCI Dataset:** We evaluated emotion recognition based on valence and arousal dimensions, with binary classification for each dimension.

For all public datasets, we adhered to the Leave-One-Subject-Out Cross-Validation (LOSO-CV) protocol. This protocol ensures that the data from one subject is used for testing while data from all other subjects are used for training. This approach not only enhances the robustness of our model but also ensures that our results are directly comparable to those reported in other state-of-the-art studies.

For our self-recorded dataset, we extract DE as the EEG modality input and a continuous sequence of face images as the micro-expression modality input. Their respective input sizes are $56 \times 56 \times 5$. The EEG modality represents 5 EEG frequency bands. The micro-expression modality represents 5 images of faces that change uniformly and continuously over 0.2 seconds. We verify the feasibility of our method on public datasets. Then, pre-training was performed on data from healthy subjects. Finally, the evaluation metrics are computed on data from patients with DoC.

In the subject-dependent experiments, we divided the training and test sets at a ratio of 4:1. In subject-independent experiments, we split all the data from the training and testing subjects at a ratio of approximately 4:1. It is also ensured that the data of the test subjects are not available during training. We use a five-fold cross-validation strategy to train the emotion recognition model. The model was implemented using the PyTorch framework and trained using the AdamW optimizer with a learning rate of $3e-4$. The learning rate was attenuated using cosine annealing. We used a batch size of 32 and trained the model for 100 epochs. We run the experiments using PyTorch libraries on an Nvidia GeForce RTX 3090 GPU.

V. RESULTS

A. Evaluation of Emotion Recognition on Public Datasets

In this section, we present a comprehensive evaluation of our novel model, the STST, across a range of public datasets, namely, the SEED [41], SEED-IV [54], CASME II [55], SAMM [56], SMIC [57], and MAHNOB-HCI [58] datasets. We begin with the SEED and SEED-IV datasets, examining the model's ability to differentiate between different emotional states based on EEG signals. Following an analysis of the combined CASME II–SAMM–SMIC dataset, the focus shifted to micro-expression recognition. Finally, the model's performance on the MAHNOB-HCI dataset, which combines both EEG and facial expression data for multimodal emotion recognition, is assessed.

We benchmarked the STST model on SEED and SEED-IV against 11 representative models, including SVM [60], DBN [61], A-LSTM [62], DGCNN [18], RGNN [63], TANN [64], BiDANN [65], BiHDM [66], MMDA-VAE [67], PGCN [68], and FGCN [69]. In the tables that follow, we first boldface our

TABLE III
COMPARISON OF EMOTION RECOGNITION ON SEED

Method (SEED)	Subject-dependent		Subject-independent	
	Acc (%)	Std (%)	Acc (%)	Std (%)
SVM [60]	83.99	9.92	56.73	16.29
DBN [61]	86.08	8.34	-	-
A-LSTM [62]	88.61	10.16	72.18	10.85
DGCNN [18]	90.40	8.49	79.95	9.02
RGNN [63]	94.24	5.95	85.30	6.72
TANN [64]	93.34	6.64	84.41	8.75
BiDANN [65]	92.38	7.04	84.14	6.87
BiHDM [66]	93.12	6.06	85.40	7.53
MMDA-VAE [67]	-	-	89.64	11.78
PGCN [68]	96.93	5.11	84.59	8.68
FGCN [69]	94.10	7.34	-	-
STST (Ours)	93.47	5.47	86.43	7.27

Bold denotes the proposed method and any competing method with equal or better performance.

own results and then additionally bold any competing result that exceeds them, so every value that is at least as high as our score is highlighted.

The comparison results of emotion recognition on the SEED dataset are shown in Table III. Our proposed STST achieved an accuracy of 93.47% with a standard deviation of 5.47% in the subject-dependent protocol and 86.43% accuracy with a standard deviation of 7.27% in the subject-independent protocol. Compared to other advanced models such as PGCN and RGNN, this approach is notably competitive. Although these models demonstrated slightly higher accuracies, they were less robust in subject-independent protocols. For instance, both the RGNN and BiHDM models showed outstanding performance in the subject-dependent protocol, but their performance significantly decreased in the subject-independent protocol. This trend highlights the difficulty of generalizing emotion recognition models across different subjects.

The comparison results of emotion recognition on the SEED-IV dataset are shown in Table IV. Our STST achieved an accuracy of 78.34% with a standard deviation of 12.41% in the subject-dependent protocol and a noteworthy 74.31% accuracy with a standard deviation of 8.42% in the subject-independent protocol. Compared to other advanced models such as PGCN and RGNN, the STST exhibits commendable performance. Although PGCN achieved the highest accuracy of 82.24% in the subject-dependent protocol, its performance in the subject-independent protocol closely matched that of the STST, indicating that our model has enhanced generalizability. RGNN and BiHDM showed promising results in the subject-dependent protocol, but their performance decreased in the subject-independent protocol, which is consistent with the trend observed in the SEED dataset analysis. There appears to be a consistent challenge across models in maintaining performance stability when transitioning from a subject-dependent to a subject-independent protocol.

We benchmarked the STST model on the combined CASME II–SAMM–SMIC dataset against 12 representative models, including LBP-TOP [70], CapsuleNet [71], OFF-ApexNet [72], DInet [73], RCN [74], ME-PLAN [75], FeatRef [76], FGRL-AUF [77], SLSTT-LSTM [6], FRL-DGT [78], SDGSA [79],

TABLE IV
COMPARISON OF EMOTION RECOGNITION ON SEED-IV

Method (SEED-IV)	Subject-dependent		Subject-independent	
	Acc (%)	Std (%)	Acc (%)	Std (%)
SVM [60]	56.61	20.05	37.99	12.52
DBN [61]	66.77	7.38	-	-
A-LSTM [62]	69.50	15.65	55.03	9.28
DGCNN [18]	69.88	16.29	52.82	-
RGNN [63]	79.37	10.54	73.84	8.02
TANN [64]	73.94	13.65	68.00	8.35
BiDANN [65]	70.29	12.63	65.59	10.39
BiHDM [66]	74.35	14.09	69.03	8.66
MMDA-VAE [67]	-	-	73.82	11.74
PGCN [68]	82.24	14.85	73.69	7.16
FGCN [69]	77.14	15.71	-	-
STST (Ours)	78.34	12.41	74.31	8.42

Bold denotes the proposed method and any competing method with equal or better performance.

and HTNet [80]. Table V presents a comparative analysis of these models' performances on the combined CASME II-SAMM-SMIC dataset, showcasing a diverse range of results. The STST model achieved a UF1 score of 0.856 and a UAR score of 0.843, placing it among the top-performing models and only marginally behind the leading models by four to six thousandths on both metrics. HTNet obtained the highest UF1 score of 0.860, while SDGSA achieved the best UAR score of 0.849.

Compared to earlier lightweight representations, the STST model demonstrated a significant improvement, with a UF1 score of 0.856, which is 0.268 higher than that of LBP-TOP. This highlights the advantage of using deep learning methods over hand-crafted descriptors. Additionally, the STST model outperformed the recent CNN-RNN hybrid SLSTT-LSTM model by 0.04 in UF1 and 0.053 in UAR. This indicates that the pure-transformer backbone of the STST model is more effective in capturing long-range micro-motion dynamics than recurrent modules.

Furthermore, the STST model maintained a balanced UF1-UAR gap of 0.013, which is smaller than the gaps of FGRL-AUF (0.002), SDGSA (0.005), and HTNet (0.012). This suggests that the STST model is less biased toward any specific emotion class and achieves stable precision and recall across all three datasets. These results underscore the model's robustness and effectiveness in micro-expression recognition.

We benchmarked the STST model on MAHNOB-HCI against 6 representative models, including NBM [81], CNN+SVM [82], CNN+LSTM [83], AM-LSTM [84], EEG-PSD+LSTM [85], and HSAttentionNet [17]. Table VI shows the results of the comparative analysis, where our STST achieved the highest accuracies in both valence (86.1%) and arousal (85.7%) detection. This superior performance highlights the robustness and efficiency of STST in handling multimodal emotional data. Our model's success is attributed to its alignment-first, modality-specific design. By maintaining separate encoders for each modality and allowing only attended heterogeneous tokens to interact in a lightweight bottleneck, STST preserves the temporal structure of EEG while effectively leveraging facial micro-expressions.

TABLE V
COMPARISON OF EMOTION RECOGNITION ON THE MERGED DATASET OF CASME II, SAMM, AND SMIC

Method	UF1	UAR
LBP-TOP [70]	0.588	0.578
CapsuleNet [71]	0.652	0.651
OFF-ApexNet [72]	0.719	0.710
DINet [73]	0.732	0.728
RCN [74]	0.743	0.719
ME-PLAN [75]	0.772	0.786
FeatRef [76]	0.784	0.783
FGRL-AUF [77]	0.791	0.793
SLSTT-LSTM [6]	0.816	0.790
FRL-DGT [78]	0.812	0.811
SDGSA [79]	0.854	0.849
HTNet [80]	0.860	0.848
STST (Ours)	0.856	0.843

Bold denotes the proposed method and any competing method with equal or better performance.

TABLE VI
COMPARISON OF EMOTION RECOGNITION ON MAHNOB-HCI

Method (MAHNOB-HCI)	Fusion Strategy	SD Acc (%)	
		Valence	Arousal
NBM [81]	Feature-level	68.5	73.0
CNN+SVM [82]	Decision-level	75.0	75.6
CNN+LSTM [83]	Feature-level	78.6	77.2
AM-LSTM [84]	Feature-level	83.2	83.8
EEG-PSD+LSTM [85]	Feature-level	85.5	82.9
HSAttentionNet [17]	Feature-level	79.9	81.4
STST (Ours)	Feature-level	86.1	85.7

SD denotes subject-dependent;

Bold denotes the proposed method and any competing method with equal or better performance.

In contrast, decision-level fusion models like CNN+SVM train independent sub-networks and fuse outputs via weighted voting. While modular, this approach cannot correct errors originating from a single modality once scores are fixed. Feature-level fusion models, such as EEG-PSD+LSTM, concatenate hand-crafted power spectra with CNN face features before temporal modeling, blurring sampling rate differences and sacrificing neurophysiological granularity.

Other feature-level fusion models, including CNN-LSTM, AM-LSTM, and HSAttentionNet, align EEG and facial features in a shared hidden space before joint classification. Most concatenate or additively fuse multimodal vectors after shared layers, risking modality dominance and temporal misalignment. By avoiding these pitfalls, STST maintains the interpretability of EEG dynamics while exploiting fine-grained facial cues.

B. Evaluation of Emotion Recognition on our Self-Recorded Dataset

This section focuses on the results obtained from our self-recorded dataset, which involves healthy subjects and patients with DoC. We explore the nuances of emotion recognition in this unique context.

Table VII shows that emotion recognition performance varied among individuals, with average accuracies of 92.64%

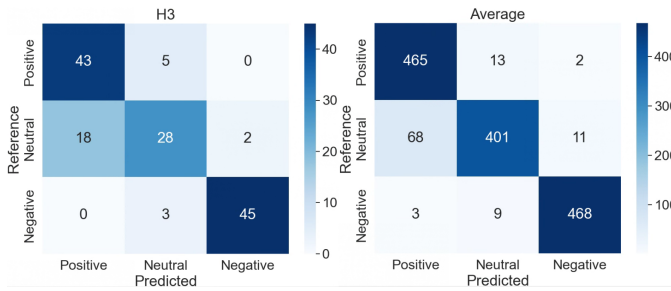


Fig. 5. Subject-dependent emotion recognition confusion matrix for healthy subjects on our self-recorded dataset.

TABLE VII
COMPARISON OF EMOTION RECOGNITION ON HEALTHY SUBJECTS IN OUR SELF-RECORDED DATASET

Healthy Subject	Subject-dependent Accuracy (%)	Subject-independent Accuracy (%)
H1	98.15	95.37
H2	95.37	91.67
H3	80.56	78.70
H4	91.67	88.89
H5	99.07	96.29
H6	97.22	84.72
H7	89.82	87.96
H8	95.37	92.59
H9	86.11	85.18
H10	93.05	91.67
Mean±Std	92.64±5.52	89.30±5.10

for the subject-dependent protocol and 89.30% for the subject-independent protocol. Notably, some healthy subjects, such as H3, exhibited lower accuracy (80.56% subject-dependent accuracy), which warrants further investigation. Figure 5 shows the confusion matrices detailing the subject-dependent emotion recognition for healthy subjects within our self-recorded dataset, suggesting a particular trend in the classification of neutral emotions. A confusion matrix for H3 reveals a tendency to misclassify neutral emotions as positive, despite high accuracy in distinguishing positive and negative emotions. This trend is less prominent in the aggregated results across 10 healthy subjects, where classification accuracy for neutral emotions improves significantly. It may reflect the inherent proximity of feature representations between neutral and positive states. This proximity may be due to the subtlety of facial expressions or physiological signals that distinguish neutral from positive states, which the model interprets as more aligned with positive emotions.

Table VIII presents Euclidean distance and cosine similarity scores comparing emotion recognition abilities between patients with DoC and healthy subjects. MCS patients generally show lower Euclidean distance and higher cosine similarity, indicating preserved emotion recognition abilities similar to healthy subjects. The marked entries indicate diagnostic changes toward the eMCS or MCS, accompanied by increases in CRS-R scores, suggesting an improvement in affective processing abilities and a possible bridging of the gap toward healthy emotional perception. In contrast, UWS patients exhibit higher Euclidean distance and significantly lower or negative cosine similarity, reflecting impaired affective

TABLE VIII
EMOTION RECOGNITION DISPARITY BETWEEN DOC PATIENTS AND HEALTHY SUBJECTS

Patient	After experiment		Metrics	
	1 month	3 months	Euclidean distance	Cosine similarity
MCS	P1* 1↑ eMCS	Rehabilitation	0.132	0.826
	P3* Rehabilitation	Rehabilitation	0.162	0.724
	P6 MCS	MCS	0.247	0.621
	P7* 1↑ MCS	8↑ eMCS	0.143	0.904
	P8 MCS	MCS	0.337	0.764
	P10* 3↑ eMCS	Rehabilitation	0.116	0.805
	P11 Not Available	Not Available	0.126	0.926
	P13* 4↑ MCS	14↑ eMCS	0.044	0.855
	P15 MCS	MCS	0.217	0.895
	P16* 5↑ eMCS	Rehabilitation	0.182	0.819
	P18 MCS	MCS	0.411	0.611
	P20† 6↑ MCS	6↑ MCS	0.073	0.824
UWS	P21 MCS	MCS	0.294	0.701
	P2 UWS	UWS	0.453	0.446
	P4 UWS	UWS	0.437	0.231
	P5 UWS	UWS	0.375	-0.423
	P9* 2↑ MCS	Not Available	0.392	0.723
	P12† 1↑ UWS	1↑ UWS	0.574	-0.817
	P14 Not Available	Not Available	0.468	-0.842
	P17 UWS	Not Available	0.378	-0.635
	P19† 2↑ UWS	2↑ UWS	0.302	0.639

Bold denotes UWS patients with improved affective processing who subsequently achieved higher CRS-R scores; Euclidean distance: [0, 2]; Cosine similarity: [-1, 1]; *: An upgrade in their level of consciousness in diagnosis after experimentation; †: Improvement in CRS-R score without an upgrade in their level of consciousness in diagnosis after experimentation; 1↑ eMCS: A one-point CRS-R score increase and transition to the eMCS after the experiment.

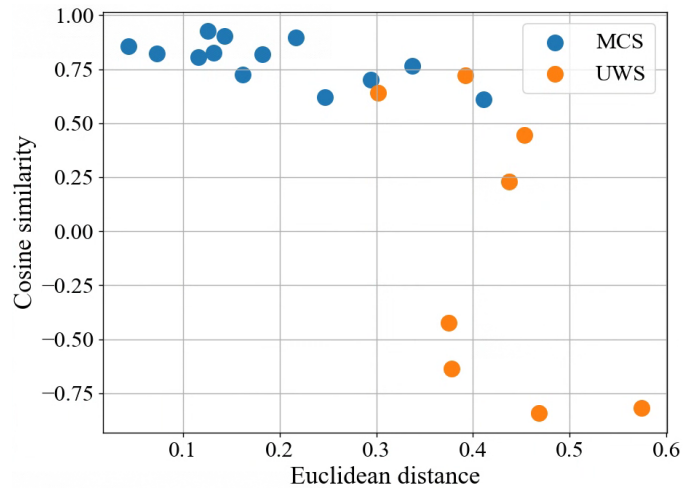


Fig. 6. Distribution of emotion recognition disparities for MCS and UWS patients.

tive resonance. Figure 6 illustrates these differences through scatter plots, highlighting the disparity in emotion recognition between MCS and UWS patients.

Descriptive statistical analysis reveals significant differences between MCS and UWS patients. MCS patients have a mean Euclidean distance of 0.212 (SD = 0.118) and a mean cosine similarity of 0.775 (SD = 0.094), while UWS patients have a

higher mean Euclidean distance of 0.420 (SD = 0.093) and a negative mean cosine similarity of -0.093 (SD = 0.610). The difference in emotional recognition capabilities between the two groups is reflected in the significant disparity in the metrics. Pearson's correlation analysis, conducted after confirming data normality, shows a significant negative correlation ($r = -0.667$, $p = 0.013$) between Euclidean distance and cosine similarity in MCS patients, indicating that greater deviation from typical emotional recognition patterns is associated with lower emotional alignment. In contrast, no significant correlation ($r = -0.486$, $p = 0.329$) is observed in UWS patients, suggesting more fragmented emotion recognition abilities. These findings highlight the complexity of emotion processing in different states of consciousness and the potential for quantitative measures to enhance diagnostic and therapeutic approaches.

VI. DISCUSSIONS

A. Ablation Study

To validate the effectiveness of the proposed spatiotemporal attention encoder (STAE) in our emotion recognition framework, we conducted an extensive ablation study. This study consisted of two main experiments designed to analyze the contributions of the STAE layers and the individual attention mechanisms they use.

Layer Ablation We varied the number of STAE layers from 0 to 5 to assess their impact on model parameters, computational cost, and predictive accuracy. The STAE integrates low-dimensional EEG features and high-dimensional micro-expression features, enhancing their representation through layer depth. Results in Table IX show that adding STAE layers significantly improves performance, with a notable jump from zero layers to one layer. This confirms the STAE's ability to enhance emotion recognition through multimodal data fusion. As the number of layers increases from 1 to 4, both subject-dependent and subject-independent accuracies improve, reaching peaks of 92.64% and 89.30%, respectively. This trend highlights the STAE's effectiveness in capturing complex emotional constructs.

However, adding a fifth layer did not provide a statistically significant improvement, with accuracies remaining at 92.71% and 89.35% for subject-dependent and subject-independent settings, respectively. This saturation point suggests that while additional layers can enhance feature extraction, they also risk overfitting and increase computational expense. This finding underscores the importance of balancing model complexity with efficiency, especially in clinical settings where quick and efficient inference is crucial. The results are visualized in the Appendix B, and the reported standard deviations indicate consistent model performance across subjects and runs, reinforcing the robustness of our framework.

Mechanistic Ablation To investigate the contributions of temporal and spatial attention mechanisms to our emotion recognition framework, we conducted an ablation study by selectively removing these modules. The goal was to assess their individual impacts on model performance and determine their importance in capturing emotion-related information.

TABLE IX
IMPACT OF STAE LAYER QUANTITY ON EMOTION RECOGNITION ACCURACY IN HEALTHY SUBJECTS

STAE Layers	Model complexity Params	FLOPs	Subject-dependent Acc±Std (%)	Subject-independent Acc±Std (%)
0	-	-	83.73±6.82	81.02±6.63
1	7.351K	18.821M	88.23±6.43	84.52±5.73
2	14.702K	37.642M	90.04±5.35	87.13±6.11
3	22.053K	56.463M	91.54±5.81	88.56±5.28
4	29.404K	75.284M	92.64±5.52	89.30±5.10
5	36.755K	94.105M	92.71±5.82	89.35±5.02

Bold denotes the 4-layer configuration that offers the best accuracy-efficiency trade-off.

TABLE X
IMPACT OF SPATIOTEMPORAL ATTENTION ON EMOTION RECOGNITION ACCURACY IN HEALTHY SUBJECTS

Attention mechanisms		Subject-dependent Acc±Std (%)	Subject-independent Acc±Std (%)
Temporal	Spatial		
-	-	83.73±6.82	81.02±6.63
✓	-	85.35±5.98	83.17±5.74
-	✓	90.07±5.32	87.23±6.09
✓	✓	92.64±5.52	89.30±5.10

Bold denotes the highest accuracy achieved when both temporal and spatial attention mechanisms are active;
The model includes 4 STAE layers, with each layer removing the corresponding attention mechanism.

Table X shows that both temporal and spatial attention significantly enhance the model's accuracy. When only temporal attention is included (without spatial attention), the subject-dependent accuracy increases from 83.73% to 85.35%, and the subject-independent accuracy rises from 81.02% to 83.17%. When only spatial attention is included (without temporal attention), the subject-dependent accuracy further improves to 90.07%, and the subject-independent accuracy reaches 87.23%. The highest accuracy is achieved when both attention mechanisms are present: 92.64% subject-dependent and 89.30% subject-independent.

These findings demonstrate the significance of both temporal and spatial attention in enhancing the model's performance. Spatial attention exhibits a more pronounced effect, as evidenced by the larger improvements in accuracy compared to the absence of spatial attention. This suggests that spatial cues, such as facial expressions, play a crucial role in capturing emotions and should receive particular attention in model design. Furthermore, spatial attention enables the model to focus on the correlations between different channels in EEG data and facial expression variations in micro-expressions, thereby facilitating a more comprehensive understanding of emotional constructs. The combination of temporal and spatial attention mechanisms results in enhanced accuracy. This suggests that temporal attention captures the temporal dynamics in EEG and micro-expression changes, while spatial attention captures the spatial patterns and facial cues associated with different emotions.

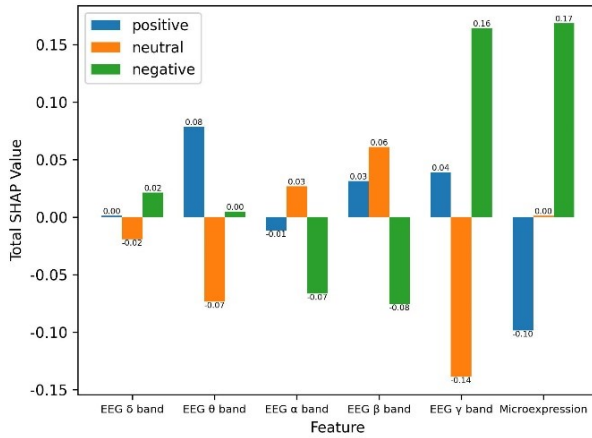


Fig. 7. Comparative Analysis of Total SHAP Values for EEG Bands and Micro-expressions Across Emotional Categories.

TABLE XI
TOTAL SHAP VALUE IMPACTS OF EEG BANDS AND MICRO-EXPRESSIONS ACROSS THREE EMOTIONAL CATEGORIES

Feature	Emotion		
	Positive	Neutral	Negative
EEG δ band	0.0012	-0.0192	0.0212
EEG θ band	0.0785	-0.0736	0.0049
EEG α band	-0.0117	0.0269	-0.0666
EEG β band	0.0312	0.0609	-0.0755
EEG γ band	0.0390	-0.1388	0.1643
Micro-expression	-0.0985	0.0012	0.1684

The values displayed in the table represent the cumulative SHAP values for each feature across a range of emotion recognition tasks. A positive value indicates a positive impact, while a negative value indicates a negative impact.

B. Interpretability

This section presents a comprehensive analysis of the roles played by EEG bands and micro-expression features in emotion recognition. It utilizes Shapley Additive exPlanations (SHAP) and EEG topographical mapping to elucidate their influence on model predictions. This analysis integrates advanced visualization techniques to facilitate a deeper understanding of how these features contribute to the classification of different emotional states.

We used SHAP analysis to quantify and visualize the contributions of five EEG bands and micro-expressions across three emotional categories: positive, neutral, and negative. SHAP values were visualized through heatmaps, showing the spatial distribution and impact magnitude of each feature. Each heatmap corresponds to one feature, displayed on a 56×56 grid, with color gradients indicating the range from negative to positive contributions (blue to red). These visualizations are detailed in Appendix C, where color gradients from blue to red indicate the range from negative to positive contributions, respectively.

Table XI and Figure 7 summarize the SHAP values for different EEG bands and micro-expressions. The EEG θ band

significantly enhanced positive emotion recognition (SHAP value of 0.078) and negatively impacted negative emotion recognition (-0.073). In contrast, the γ band played a crucial role in recognizing negative emotions (SHAP value of 0.164). Micro-expressions also strongly contributed to negative emotion classification (SHAP value of 0.168). These findings highlight the distinct roles of EEG bands and micro-expressions in emotion recognition. The distinct roles of EEG bands suggest that different frequencies are tuned to specific emotional resonances. The θ and γ bands are pivotal in distinguishing between positive and negative emotional states. Moreover, the significant role of micro-expressions in identifying negative emotions is consistent with psychological theories that emphasize the salience of facial expressions in conveying distress. Furthermore, it has been demonstrated that the stable pattern of brain electricity is consistent across different experiments. The activation of positive emotions was greater than that of negative emotions in the θ and β bands. Furthermore, a higher response was observed in the δ and γ bands for negative emotions. This finding is consistent with the conclusions of previous studies [41].

EEG topographical maps further illustrate the relationship between EEG channels and emotional states. Figure 8 compares EEG power spectral density maps for healthy subjects, MCS patients, and UWS patients while viewing negative emotion videos. The topographical maps of the healthy subjects revealed greater power responses in the parietal and frontal regions for negative emotions, beginning with the δ band. Similarly, a comparable pattern was observed in the MCS patients. The δ band model observed in UWS patients suggests the potential for cognitive processing deficits. When the γ band was examined, the healthy subjects exhibited a consistent and robust distribution of power, with the greatest concentration observed in the prefrontal regions. In contrast, the MCS patient was more similar to the healthy subject, whereas the UWS patient showed reduced power and fewer localized patterns in all of these frequency bands. δ waves are typically associated with deep sleep, and a greater number of δ waves and a lower number of other high-frequency waves may indicate that the individual is in a state of deep relaxation or sleep. UWS patients exhibit reduced power in all frequency bands, with the exception of δ . This finding suggests that patients are not aware of their surroundings and have difficulty distinguishing reality.

These analyses provide valuable insights into the neural processes underlying emotion recognition. The SHAP values offer a detailed understanding of feature importance, while EEG topographical maps reveal distinct brain activity patterns across different cognitive states. These findings can inform the development of more effective diagnostic tools for neurological disorders, including emotion recognition and assisted diagnosis of DoC.

C. Limitations and Future Work

This study presents a novel approach to emotion recognition in patients with DoC by integrating EEG and micro-expression data. However, several limitations exist. First, deploying our

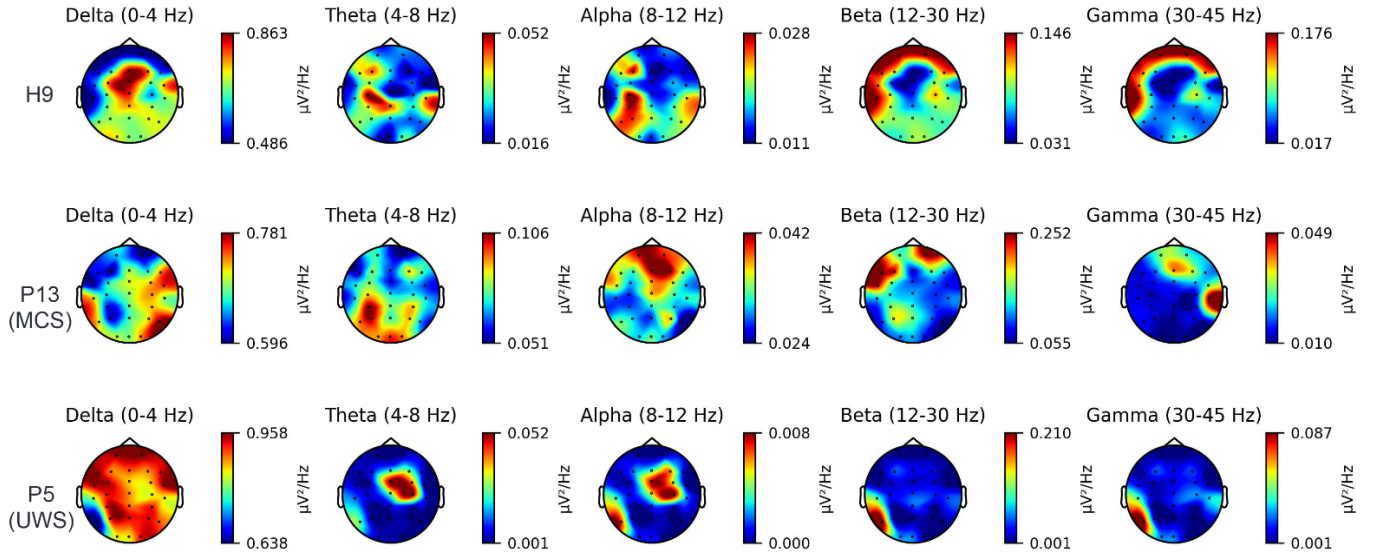


Fig. 8. EEG power spectral density topographical maps of different frequency bands during negative emotion video viewing for healthy, MCS, and UWS subjects.

computational models in real-time clinical settings requires significant resources, which may be impractical in resource-limited environments. Additionally, the spatio-temporal attention mechanisms used for emotion recognition are still developing. While effective for normal individuals, their accuracy and reliability in various clinical conditions remain uncertain. The integration of this technology into existing healthcare workflows and its practical applicability in routine clinical operations also need further exploration.

Future work should focus on refining and validating the proposed model to enhance its clinical and economic feasibility. Developing cost-effective and efficient models is essential for broader accessibility and integration into medical systems. Extensive clinical trials are needed to confirm the efficacy, usability, and integration of the emotion recognition system in real-world settings. Additionally, future research should aim to improve the stability and accuracy of the algorithms across diverse patient demographics and clinical conditions.

We are committed to making our dataset publicly available in the near future. This initiative will significantly support further research in this field and facilitate the application of artificial intelligence in medical diagnostics. By making this dataset public, we aim to attract a broader research community to explore this subfield, thereby accelerating advancements in the auxiliary diagnosis of disorders of consciousness. We believe that the availability of this dataset will contribute to more robust and innovative solutions in the medical field.

VII. CONCLUSION

This study presents a multimodal framework for emotion recognition that introduces an attention-based heterogeneous-fusion module and a clinically grounded evaluation paradigm. By integrating EEG and micro-expression analyses, this framework assesses emotional states in patients with DoC. Our find-

ings reveal that this integrative method significantly enhances the accuracy of emotion recognition. It achieves comparable accuracy to mainstream models on the EEG datasets SEED and SEED-IV, as well as the micro-expression combined CASME II-SAMM-SMIC dataset. State-of-the-art results were achieved on the multimodal emotion recognition dataset MAHNOB-HCI. In our self-recorded dataset, the accuracy of subject-dependent and subject-independent emotion recognition was $92.64 \pm 5.52\%$ and $89.30 \pm 5.10\%$, respectively, for healthy subjects.

The principal contribution of our work is the development and validation of a spatiotemporal attention mechanism that effectively fuses data from multiple modalities, thereby enabling the capture of subtle emotional expressions in patients with DoC, who are often challenging to assess using conventional methods. This approach not only enhances diagnostic precision but also facilitates a better understanding of emotional nuances in patients with severely impaired consciousness. The implications of our research are profound, offering the potential for enhancements in clinical practice for monitoring and managing patients with DoC. The capacity to accurately recognize emotions in these patients can facilitate the implementation of more personalized care strategies and enhanced outcomes. This is achieved by assisting clinicians in making informed decisions regarding patient consciousness levels and potential recovery pathways. This study highlights the necessity for continued research into optimizing multimodal emotion recognition systems for practical and widespread clinical application. We are committed to making our dataset publicly available in the future to support further research in this field. This initiative will significantly support the application of artificial intelligence in medical diagnostics and advance related research. Furthermore, the research could be expanded to investigate the scalability of the approach and

its applicability to other neurological conditions beyond DoC, thus broadening the scope of its clinical utility. The continued development and integration of such technologies has the potential to transform clinical practices and enhance patient care for some of the most vulnerable populations.

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